

EmPay

Smart Human Resource Management System

Odoo Hackathon 2026 · Team Presentation

Frontend Dev · Backend Dev · Backend Support · **Project Explainer**

The Problem

Fragmented tools

Attendance, payroll, and leave tracked in silos — spreadsheets, paper registers, separate apps.

Error-prone payroll

Manual salary calculations lead to costly mistakes, especially with PF, professional tax, and leave deductions.

No role clarity

Sensitive payroll data visible to HR; employees can't self-serve their own attendance or payslips.

Zero audit trail

No locking mechanism — finalized payslips can be edited, creating compliance risk.

EmPay — The Solution

One unified platform. Four powerful modules.

01

User & Role Management

- JWT Auth
- RBAC
- Profile Mgmt

02

Attendance & Leave

- Check-in/out
- Leave workflow
- Balances

03

Payroll Engine

- PF + Prof Tax
- Payslips
- Locked runs

04

Dashboard & Analytics

- Charts
- PDF reports
- Role views

Who Uses EmPay?

Admin

Full access

- Manage all users & roles
- All modules — no limits
- System settings & config

HR Officer

People & Attendance

- Create/update employee profiles
- Monitor all attendance
- Allocate leave balances

Payroll Officer

Compensation & Reports

- Approve/reject time-off
- Generate payslips & reports
- View attendance (read-only)

Employee

Self-Service

- Mark daily attendance
- Apply for leave
- View own payslip & records

How EmPay Works

The dependency chain that drives accurate payroll

- 1 Employee record**
Created first — department, designation, basic salary, join date.
- 2 Attendance tracked**
Daily check-in/out auto-calculates hours worked and status (Present / Absent / Half-Day).
- 3 Leave approved**
Officer approves/rejects requests. Approved leaves adjust the working-days count.
- 4 Payrun triggered**
Attendance + leaves pulled for the pay period. Engine calculates Gross, PF (12%), Prof Tax, Net Pay.
- 5 Payslip locked**
Post-finalization, payslips are auto-generated, locked, and available for employee download.

Payroll Calculation

Salary Breakdown

Basic Salary	Base pay
+ HRA	House rent allowance
+ Allowances	Travel, medical, etc.
= Gross Salary	Sum of above
– PF Deduction	12% of Basic
– Professional Tax	State-levied
– Other Deductions	If applicable
= Net Pay	Employee receives

12%

PF (of Basic)

₹200

Prof Tax/mo

5 steps Locked

Payrun process

Post-finalize

Once finalized, payslips are locked and downloadable as PDF. Audit trail is maintained automatically.

Tech Stack & Architecture

Frontend

React / Next.js · Tailwind CSS · Axios · React Query · Chart.js

Backend API

Node.js + Express.js · JWT + bcrypt · RBAC middleware · REST endpoints

Database

PostgreSQL / MySQL · Prisma / Sequelize ORM · 8 relational tables

DevOps & Reports

Docker (optional) · PDFKit / Puppeteer · Jest · GitHub Actions CI

3-tier architecture · RBAC on every route · <2s response time SLA · JWT session management

Demo Walkthrough

Four roles — one end-to-end story

Admin

Register, assign roles, view full dashboard

HR Officer

Create employee, allocate 12 annual + 6 sick leaves, view attendance

Employee

Check-in, apply for leave, view payslip

Payroll Officer

Approve leave, run payroll, generate & lock payslips, export PDF

Team & Responsibilities

M1

Frontend Developer

- All UI screens & role-aware navigation
- React / Next.js + Tailwind CSS
- Chart.js dashboards & form validation

M2

Backend Lead

- DB schema, Auth APIs, RBAC
- Employee, Attendance, Leave, Payroll APIs
- Payroll engine & Payslip logic

M3

Backend Support

- Analytics APIs + PDF reports
- Data seeding & Jest unit tests
- Docker setup & Git management

M4

Project Explainer

- SRS document & architecture diagram
- Presentation, demo script & README
- UAT walkthroughs & hackathon demo lead

Thank You

EmPay — Smarter payroll starts with better systems.